

Laplace Transforms

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TABLE 12.1 **An Abbreviated List of Laplace Transform Pairs**

Type	$f(t)$ ($t > 0-$)	$F(s)$
(impulse)	$\delta(t)$	1
(step)	$u(t)$	$\frac{1}{s}$
(ramp)	t	$\frac{1}{s^2}$
(exponential)	e^{-at}	$\frac{1}{s + a}$
(sine)	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
(cosine)	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
(damped ramp)	te^{-at}	$\frac{1}{(s + a)^2}$
(damped sine)	$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
(damped cosine)	$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

TABLE 12.2 An Abbreviated List of Operational Transforms

Operation	$f(t)$	$F(s)$
Multiplication by a constant	$Kf(t)$	$KF(s)$
Addition/subtraction	$f_1(t) + f_2(t) - f_3(t) + \cdots$	$F_1(s) + F_2(s) - F_3(s) + \cdots$
First derivative (time)	$\frac{df(t)}{dt}$	$sF(s) - f(0^-)$
Second derivative (time)	$\frac{d^2f(t)}{dt^2}$	$s^2F(s) - sf(0^-) - \frac{df(0^-)}{dt}$
n th derivative (time)	$\frac{d^nf(t)}{dt^n}$	$s^nF(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt} - s^{n-3}\frac{df^2(0^-)}{dt^2} - \cdots - \frac{d^{n-1}f(0^-)}{dt^{n-1}}$
Time integral	$\int_0^t f(x) dx$	$\frac{F(s)}{s}$
Translation in time	$f(t-a)u(t-a), a > 0$	$e^{-as}F(s)$
Translation in frequency	$e^{-at}f(t)$	$F(s+a)$
Scale changing	$f(at), a > 0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
First derivative (s)	$tf(t)$	$-\frac{dF(s)}{ds}$
n th derivative (s)	$t^n f(t)$	$(-1)^n \frac{d^n F(s)}{ds^n}$
s integral	$\frac{f(t)}{t}$	$\int_s^\infty F(u) du$

Partial Fraction Expansion: Distinct Real Roots of $D(s)$

$$F(s) = \frac{96(s + 5)(s + 12)}{s(s + 8)(s + 6)} \equiv \frac{K_1}{s} + \frac{K_2}{s + 8} + \frac{K_3}{s + 6}.$$

$$\left. \frac{96(s + 5)(s + 12)}{(s + 8)(s + 6)} \right|_{s=0} \equiv K_1 + \left. \frac{K_2 s}{s + 8} \right|_{s=0} + \left. \frac{K_3 s}{s + 6} \right|_{s=0},$$

$$\frac{96(5)(12)}{8(6)} \equiv K_1 = 120.$$

Final expression and check at $s = -5$

$$\frac{96(s+5)(s+12)}{s(s+8)(s+6)} \equiv \frac{120}{s} + \frac{48}{s+6} - \frac{72}{s+8}.$$

$$\frac{120}{-5} + \frac{48}{1} - \frac{72}{3} = -24 + 48 - 24 = 0,$$

Partial Fraction Expansion: Distinct Complex
Roots of $D(s)$

$$F(s) = \frac{100(s + 3)}{(s + 6)(s^2 + 6s + 25)}.$$

$$s^2 + 6s + 25 = (s + 3 - j4)(s + 3 + j4).$$

$$\frac{100(s + 3)}{(s + 6)(s^2 + 6s + 25)} \equiv$$

$$\frac{K_1}{s + 6} + \frac{K_2}{s + 3 - j4} + \frac{K_3}{s + 3 + j4}.$$

$$K_1 = \left. \frac{100(s + 3)}{s^2 + 6s + 25} \right|_{s=-6} = \frac{100(-3)}{25} = -12,$$

$$\begin{aligned}
 K_2 &= \left. \frac{100(s+3)}{(s+6)(s+3+j4)} \right|_{s=-3+j4} = \frac{100(-j4)}{(3+j4)(j8)} \\
 &= 6-j8 = 10e^{-j53.13^\circ},
 \end{aligned}$$

$$\begin{aligned}
 K_3 &= \left. \frac{100(s+3)}{(s+6)(s+3-j4)} \right|_{s=-3-j4} = \frac{100(-j4)}{(3-j4)(-j8)} \\
 &= 6+j8 = 10e^{j53.13^\circ}.
 \end{aligned}$$

$$\frac{100(s + 3)}{(s + 6)(s^2 + 6s + 25)} = \frac{-12}{s + 6} + \frac{10 \angle -53.13^\circ}{s + 3 - j4}$$

$$+ \frac{10 \angle 53.13^\circ}{s + 3 + j4}.$$

Testing at -3

$$F(s) = \frac{-12}{3} + \frac{10 \angle -53.13^\circ}{-j4} + \frac{10 \angle 53.13^\circ}{j4}$$

$$= -4 + 2.5 \angle 36.87^\circ + 2.5 \angle -36.87^\circ$$

$$= -4 + 2.0 + j1.5 + 2.0 - j1.5 = 0.$$

$$\mathcal{L}^{-1} \left\{ \frac{100(s + 3)}{(s + 6)(s^2 + 6s + 25)} \right\} = (-12e^{-6t} + 10e^{-j53.13^\circ} e^{-(3-j4)t} \\ + 10e^{j53.13^\circ} e^{-(3+j4)t})u(t). \quad ($$

$$\begin{aligned}
& 10e^{-j53.13^\circ}e^{-(3-j4)t} + 10e^{j53.13^\circ}e^{-(3+j4)t} \\
&= 10e^{-3t}(e^{j(4t-53.13^\circ)} + e^{-j(4t-53.13^\circ)}) \\
&= 20e^{-3t}\cos(4t - 53.13^\circ),
\end{aligned}$$

$$\begin{aligned}
& \mathcal{L}^{-1}\left\{\frac{100(s+3)}{(s+6)(s^2+6s+25)}\right\} \\
&= [-12e^{-6t} + 20e^{-3t}\cos(4t - 53.13^\circ)]u(t).
\end{aligned}$$

As we saw in last example, k_2 and k_3 are conjugate of each other, therefor

$$\frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta}$$

$$K = |K|e^{j\theta} = |K|\angle\theta^\circ,$$

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta} \right\} \\ = 2|K|e^{-\alpha t} \cos(\beta t + \theta). \end{aligned}$$

Poles and zeros

$$F(s) = \frac{K(s + z_1)(s + z_2) \cdots (s + z_n)}{(s + p_1)(s + p_2) \cdots (s + p_m)},$$

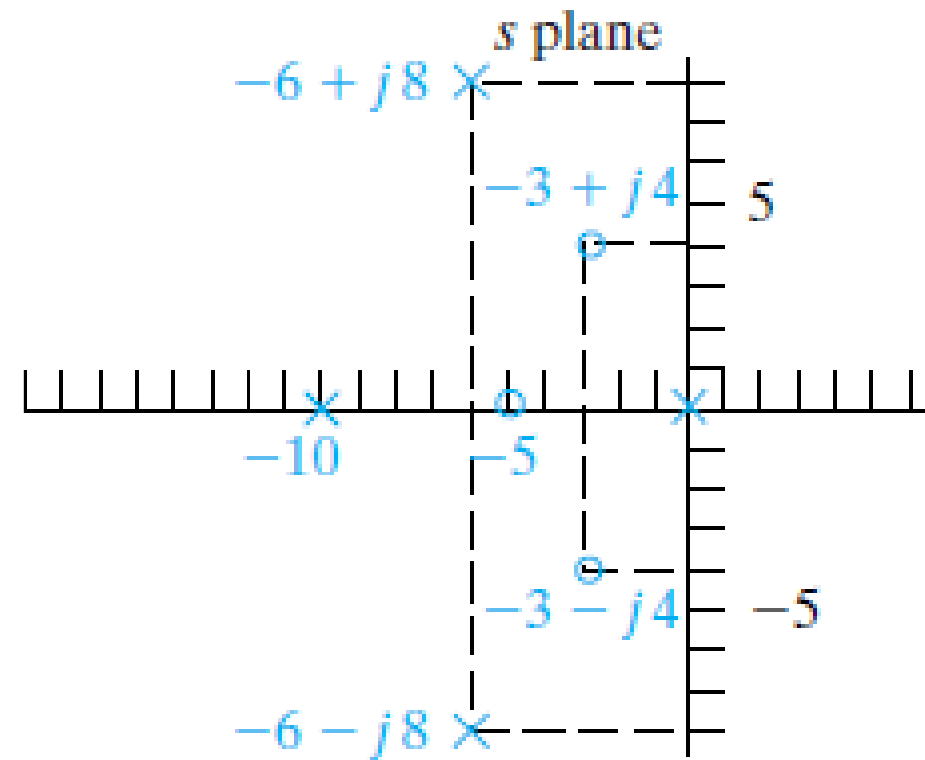
$$\begin{aligned} F(s) &= \frac{8(s^2 + 15s + 50)}{2(s^4 + 10s^3 + 35s^2 + 50s + 24)} \\ &= \frac{4(s + 5)(s + 10)}{(s + 1)(s + 2)(s + 3)(s + 4)}. \end{aligned}$$

Poles are -1, -2, -3, -4

Zeros are -5, -10

X represent poles, o represents zeros

$$F(s) = \frac{10(s + 5)(s + 3 - j4)(s + 3 + j4)}{s(s + 10)(s + 6 - j8)(s + 6 + j8)}$$



Assessment problems upto 12.8, plus related
end problems